Tab 1

PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)

100-ft Ring Road, Bengaluru – 560 085, Karnataka, India

Dissertation on

‘ EMG based control of a Robotic Arm using Deep Learning’

Submitted by

Nihal Nidhi Shivam	**(PES1UG23EE033)**
Sanjana S	**(PES1UG23EE044)**
Sanketh M	**(PES1UG23EE045)**
Srihari S	**(PES1UG23EE052)**

Aug. - Dec. 2025

under the guidance of

Guide	***Chairperson***
Prof. PUSHPA K R **Department of EEE** **PES University** **Bengaluru -560085**	**Dr. REX JOSEPH** **Department of EEE** **PES University** **Bengaluru -560085**

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

BACHELOR OF TECHNOLOGY

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

PROGRAM - Bachelor of Technology

CERTIFICATE

This is to certify that the Dissertation entitled

‘EMG based Control of a Robotic Arm using Deep Learning’

is a bonafide work carried out by

Nihal Nidhi Shivam	**(PES1UG23EE033)**
Sanjana S	**(PES1UG23EE044)**
Sanketh M	**(PES1UG23EE045)**
Srihari S	**(PES1UG23EE052)**

In partial fulfillment for the completion of **5th** semester capstone project phase -1 (UE23EE321A) in the Program of Study Bachelor of Technology in **Electrical and Electronics Engineering** under rules and regulations of PES University, Bengaluru during the period Aug. 2025 – Dec. 2025. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The dissertation has been approved as it satisfies the 5th semester academic requirements in respect of project work

Signature with date & Seal	***Signature with date & Seal***	***Signature with date & Seal***
Guide	*Chairperson*	*Dean of Research*
Prof.Pushpa K R (Assistant Professor)	*Dr.Rex Joseph*	*Dr.J.Manikandan*
Electrical and Electronics Engineering	*Electrical and Electronics Engineering*	*Electrical and Electronics Engineering*

DECLARATION

We, **Nihal Nidhi Shivam, Sanjana S, Sanketh M and Srihari S,** hereby declare that the dissertation entitled, ***'EMG based control of a robotic arm using deep learning'****,* is an original work done by us under the guidance of **Prof. Ms. Pushpa KR** ,Associate Professor and is being submitted in partial fulfillment of the requirements for completion of 5th Semester course work in the Program of Study Bachelor of Technology. in Electrical and Electronics Engineering.

PLACE: Bengaluru

DATE: 21-11-25

NAME AND SIGNATURE OF THE CANDIDATE

Nihal Nidhi Shivam
Sanjana S
Sanketh M
Srihari S

ACKNOWLEDGMENT

We would like to express our sincere gratitude to the respected dignitaries of PES University for their constant support and encouragement for carrying out this project work.

We pay our heartfelt tribute to the late ***Dr. M.R. Doreswamy*** *(Founder, PES University), whose visionary leadership and support laid the foundation for a nurturing academic environment. His contributions will always be remembered with deep respect and gratitude.*

We also extend our deep sense of gratitude to ***Prof. Jawahar Doreswamy****, the Chancellor of PES University and* ***Dr. J Surya Prasad****, the Vice Chancellor of PES University for giving us an opportunity to be a student of this reputed institution.*

We extend our respect to ***Dr. K.S. Sridhar****, Registrar of PES University for his valuable support to conduct this Project under PES University.*

It is our privilege to thank the Chairperson of the Department ***Dr. Rex Joseph****, department of Electrical and Electronics Engineering for his support and guidance for doing our Project.*

We express our sincere gratitude to our guide ***Prof.Pushpa K P*** *for her valuable guidance and suggestion, technically and logically for doing our Project work.*

We also express our gratitude to all our faculty members, parents and fellow mates who have helped us to carry out this work. Last but not the least, we thank almighty God for his blessings showered on us during this Project period.

ABSTRACT

Surface Electromyography (sEMG) provides a fast and non-invasive way to capture muscle activity, enabling intuitive human–machine interaction. This project presents an EMG-based control system designed to recognize user gestures and map them to robotic arm movements in real time. Muscle signals are acquired using BioAmp sensors and digitized through a DAQ interface. The raw EMG data undergoes preprocessing, including Butterworth filtering, rectification, RMS envelope extraction, and normalization to ensure high-quality signal representation. A set of time-domain and frequency-domain features is computed from each window of the processed signal, forming the input to a deep learning classifier trained for gesture recognition. The predicted gestures are then translated into corresponding robotic arm actions, allowing smooth and responsive control. The system demonstrates a practical approach for prosthetic applications, rehabilitation, assistive devices, and human–machine interfaces, showing how physiological signals can be effectively leveraged for intelligent, user-driven control.

TABLE OF CONTENTS

CHAPTER 1 : Introduction

Electromyography (EMG) provides a direct window into muscle activation patterns, enabling precise interpretation of human intent for applications in biomechanics, rehabilitation, and next-generation robotic control. Recent advances in biomedical hardware and signal processing have enabled the reliable acquisition and analysis of EMG signals in dynamic, real-world environments. Nevertheless, challenges remain due to the presence of motion artifacts, electrical noise, and variability across users and recording conditions.

Motivated by current literature and the emerging need for robust human-machine interfaces, this project focuses on developing an advanced EMG-based gesture recognition pipeline tailored for real-time robotic control. The system integrates high-fidelity BioAmp and DAQ hardware with rigorous digital preprocessing and machine learning algorithms, ensuring accurate feature extraction and rapid, noise-tolerant classification. By emphasizing adaptive signal processing, dataset diversity, and low-latency response, the proposed solution addresses key requirements for interactive robotics, prosthetic control, and clinical assessment, laying a foundation for future intelligent systems driven by physiological signals.

1.1. Background

The evolution of Human-Computer Interaction (HCI) is rapidly shifting from traditional mechanical inputs—such as joysticks, keyboards, and buttons—toward more intuitive, bio-signal-based interfaces. Among these, Surface Electromyography (sEMG) stands out as a powerful modality for bridging the gap between human intent and machine response. sEMG measures the electrical activity generated by muscle motor units during contraction, providing a direct window into the user's physiological commands without the need for invasive surgical procedures.

1.2. Overview

The field of biomedical engineering has rapidly evolved with advancements in artificial intelligence, wearable sensing, and signal processing, and among these innovations, surface electromyography (sEMG) has become an essential tool for understanding human muscle activity. sEMG captures the tiny electrical signals generated during muscle contraction, enabling applications in rehabilitation, prosthetic control, sports performance, and human–machine interaction. However, despite the availability of modern sensors, interpreting raw EMG data in real-world conditions remains challenging because the signal is highly sensitive to noise, electrode placement, motion artifacts, power-line interference, and physiological variations between users.

This complexity goes far beyond basic filtering or visual inspection, demanding a deeper understanding of how various signal attributes interact.This project aims to bridge that gap by integrating robust preprocessing with intelligent analysis to create a practical EMG evaluation system that operates reliably on real-world data. Through this combined approach, EMG interpretation can move closer to everyday usability, offering more accurate, meaningful insights for rehabilitation, prosthetics, and interactive biomedical technologies.

This project presents a comprehensive solution designed to acquire, process, and translate sEMG signals into real-time robotic arm actuation. The system is built upon a hybrid framework of robust analog hardware and advanced Deep Learning algorithms.

1.3. Problem Definition

The accurate replication of EMG signals in real-world robotic and gesture recognition systems remains a major technical hurdle, especially under conditions of dynamic motion. Most existing EMG pipelines are developed and tested in controlled environments, resulting in limited robustness when exposed to natural signal variability.

A key limitation in EMG research is that systems treat noise and signal features separately, failing to handle their combined effects during real-world movement. Baseline drift, crosstalk, and motion artifacts can distort the waveform and reduce accuracy, while processing delays prevent real-time control. Moreover, most datasets lack the natural variability of dynamic motion, making current models unable to generalize well outside laboratory conditions.

This project aims to overcome these limitations by constructing a unified EMG processing framework that:

1.Collects comprehensive EMG data that accurately reflects real muscle activity during dynamic movement,
2.Applies advanced noise-reduction and artifact-suppression techniques to produce a clean and stable signal,
3.Designs a low-latency processing pipeline that minimizes delay while preserving essential muscle features, and
4.Validates the system's ability to faithfully replicate EMG patterns and perform fast, noise-resistant gesture recognition under movement-rich conditions.

The challenge lies in building a pipeline that is both technically robust and capable of handling the subtle, rapidly changing characteristics of EMG signals during real-world motion. The goal is to move from basic EMG acquisition to a reliable, practical system that supports stable control and accurate recognition in dynamic environments.

CHAPTER 2 : Literature Survey

[1].This review paper summarizes major EMG signal processing methods and classification techniques.It explains noise sources, preprocessing steps, and commonly used feature extraction approaches.The work provides a strong foundation for developing accurate EMG-based control and recognition systems.

[2].This paper evaluates the first-difference transformation for extracting features from EMG time-series data.It shows that this method enhances class separability and reduces noise effects in EMG signals.The approach improves the accuracy of EMG-based pattern recognition systems

[3].This paper reviews the fundamentals of surface EMG and its role in biomechanics.It explains how EMG signals reflect muscle activation and the factors affecting their accuracy.The work provides foundational guidelines for collecting and interpreting EMG data in research and clinical applications.

[4].This paper analyzes key factors affecting EMG signal quality, including sampling rate, noise reduction, and amplitude estimation.It highlights how improper processing can distort physiological information.The study provides guidelines for optimal EMG acquisition and preprocessing for accurate analysis.

[5].This study evaluates mean and median frequency features for classifying EMG signals.Results show that frequency-domain features effectively differentiate muscle activities and movement types.The paper supports their use in EMG-based pattern recognition and gesture-control systems.

[6].This survey reviews the architecture and methods used in myoelectric control systems.It covers signal acquisition, feature extraction, and classification techniques for EMG-based control.The paper highlights challenges and provides design guidelines for prosthetic and human–machine interface applications.

[7].This paper reviews major EMG signal processing steps including filtering, feature extraction, and classification.It explains common noise sources and compares different preprocessing and recognition methods.The study provides a comprehensive foundation for designing accurate EMG-based control and analysis systems.

[8].The paper develops a deep learning model to estimate finger angles from EMG signals.It uses attention to highlight important muscle features for prediction.The approach improves accuracy while offering explainable insights for EMG-based control.

CHAPTER 3 : EMG Signal Acquisition and Preprocessing

3.1 Introduction

Surface Electromyography (sEMG) measures the electrical activity generated by muscle fibers during voluntary contraction. The raw EMG signal is extremely weak, random, and highly susceptible to noise, making preprocessing essential prior to its use in biomedical applications such as prosthetics, rehabilitation robotics, and human–machine interaction.
As noted in the literature, EMG signals contain rich neuromuscular information but are severely contaminated by various noise sources, necessitating robust signal conditioning and processing .

3.2 Characteristics of Raw EMG Signals

3.2.1 Amplitude and Frequency Characteristics

- Amplitude: **10 µV – 2 mV**
- Frequency Range: **20 – 450 Hz**

3.2.2 Nature of the Signal

- Non-stationary
- Bipolar
- Sensitive to electrode placement and skin–electrode impedance

3.2.3 Implications

Due to its low amplitude, the EMG signal is extremely vulnerable to external and physiological noise, emphasizing the need for amplification and filtering prior to analysis .

3.3 Noise Sources in EMG Acquisition

3.3.1 Power-Line Interference (PLI)

- 50/60 Hz hum from surrounding electrical equipment.

3.3.2 Baseline Wandering

- Slow drift due to respiration or limb movement.

3.3.3 White Gaussian Noise

- Random electronic noise from amplifiers and sensors.

3.3.4 Motion Artifacts

- Sudden spikes caused by electrode displacement.

3.3.5 Cross-Talk

- EMG picked up from adjacent muscles.

These noise sources degrade the signal-to-noise ratio (SNR) and reduce classification accuracy for prosthetic control and biomedical analysis .

3.4 Need for Preprocessing

Preprocessing is required to:

- Remove high- and low-frequency noise
- Enhance SNR
- Retain true muscle activation patterns
- Improve the reliability of downstream algorithms (feature extraction, classification, pattern recognition)

A robust preprocessing pipeline ensures high-quality EMG suitable for research and real-time control applications .

3.5 Data Acquisition System

3.5.1 Role of the Bio-Amplifier (BIOAMP)

A biomedical amplifier is essential for boosting very weak EMG signals and preparing them for digitization.

Stages of Bio-Amplifier Processing

(a) Differential Amplification

- Measures potential difference between electrodes.
- High CMRR (>80 dB) rejects common noise.

(b) High Gain Amplification

- Raw EMG (10 µV–5 mV) is amplified by **×1000 to ×10,000**.

- Ensures compatibility with ADC input range.

(c) Filtering (Hardware Conditioning)

- **High-pass filter (10–20 Hz):** Removes motion artifacts
- **Low-pass filter (450–500 Hz):** Eliminates high-frequency noise
- **Notch filter (50/60 Hz):** Removes power-line interference

(d) Impedance Matching

- Minimizes signal loss and distortion.

(e) Electrical Isolation

- Protects the user from potential hazards.

(f) Output

- Conditioned analog EMG is fed to **ADC → MATLAB/microcontroller** for further processing.

These conditioning stages are emphasized in literature as critical for noise reduction and accurate EMG interpretation .

3.6 Digitization Methods

Once amplified and filtered, the EMG signal is digitized using:

3.6.1 MATLAB-Based DAQ Systems

- 16–24-bit accuracy
- Sampling: 1–10 kHz
- High precision, but expensive

3.6.2 Arduino (10-bit ADC)

- Sampling: 500–1000 Hz
- Economical option for student projects

- Lower resolution compared to DAQ

3.6.3 Hybrid Systems

- DAQ for recording EMG
- Arduino for actuator/robotic control
- Provides flexibility with increased complexity

3.6.4 Wireless EMG Systems

- EMG → Microcontroller → Bluetooth → PC/mobile
- Suitable for wearable devices
- Limited by bandwidth/latency

3.7 Relevance to Research Literature

Studies highlight:

- The importance of preprocessing to preserve motor unit action potentials (MUAPs)
- Noise removal methods such as:
  - Wavelet Transform
  - Empirical Mode Decomposition (EMD/EEMD)
  - Higher Order Statistics (HOS)
  - Independent Component Analysis (ICA)
  - Neural Network-based filtering

The literature strongly supports the necessity of robust acquisition and advanced noise-removal methods to ensure high-quality EMG data for research and clinical applications .

CHAPTER 4 : EMG Signal Processing Pipeline: From Butterworth Filtering to Feature Extraction

4.1. Introduction

Surface electromyography (EMG) records electrical activity from skeletal muscles using surface electrodes placed on the skin. EMG signals provide direct insight into muscle activation patterns and are widely used in biomechanics, clinical diagnostics, and control of prosthetic devices.[1]

However, raw EMG signals suffer from several issues:

- **Low-frequency drift** from electrode movement and respiration
- **High-frequency noise** from power lines and electronic equipment
- **Motion artifacts** and baseline wandering

These contaminants must be removed through systematic preprocessing to ensure reliable feature extraction and classification.[2][1]

4.2. Technical Background

4.2.1 EMG Signal Characteristics

EMG signals typically exhibit:

1. **Frequency range:** 30–300 Hz (muscle activity)
2. **Amplitude range:** Microvolts to millivolts (depending on amplification)
3. **Bipolar nature:** Signals oscillate around zero due to action potentials

Contamination sources include:

1. Motion artifacts (<20 Hz)
2. Power line interference (50/60 Hz and harmonics)
3. High-frequency electronic noise (>300 Hz)

4.2.2 Filter Theory and Comparison

Digital filters are essential for isolating the desired EMG frequency band. Several filter types are available, each with distinct characteristics:[2]

Table 1: Comparison of Common Filter Types Butterworth filters

Filter Type	Pros	Cons	Typical Use
Butterworth	Maximally flat passband; simple implementation	Nonlinear phase, can distort timing	EMG, most biomedical signals
Chebyshev I	Steep roll-off, lower order needed	Passband ripple, less flat response	Communications, selectivity
FIR	Linear phase (preserves timing)	Requires higher order for sharp cutoff	Audio, phase-critical applications
Bessel	True linear phase, best step response	Slow roll-off, not ideal for sharp cutoffs	Event detection, timing
Elliptic	Sharpest cutoff, adjustable ripple	Ripple in passband/stopband, complex	Selective filters

4.3. Methodology

The complete signal processing pipeline is illustrated in **Figure 1** below:

Raw EMG Signal → Bandpass Filter (30-300 Hz) → Rectification →
RMS Envelope → Normalization → Windowing → Feature Extraction → Feature Matrix

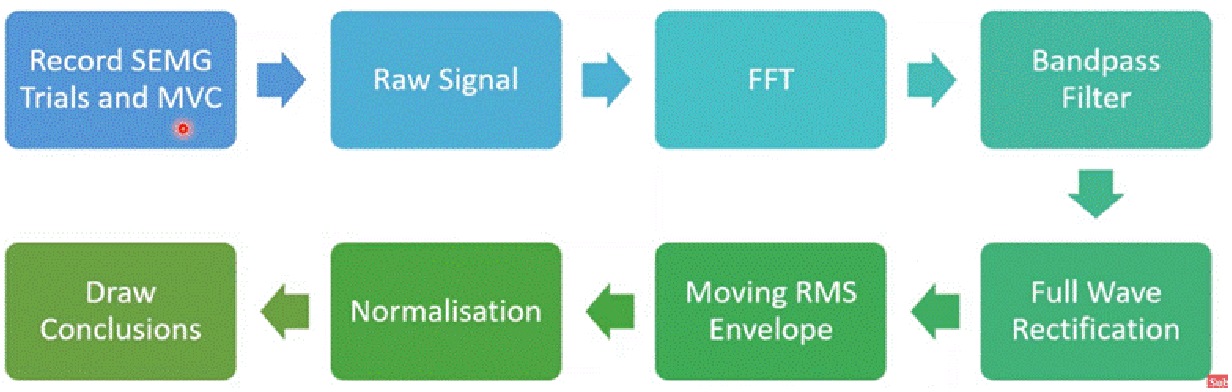

Figure 1: Block Diagram of the sEMG Signal Processing Workflow

4.3.1 Data Import and Initialization

EMG data was imported from a text file containing voltage samples acquired at a sampling frequency of 1000 Hz. This sampling rate satisfies the Nyquist criterion for capturing all relevant muscle activity frequencies up to 500 Hz.[1]

Code Snippet 1: Data Import

```
clear;
clc;
disp('--- EMG Signal Processing Started ---');

% Import EMG data
data = readmatrix('emg_healthy.txt');
disp(['Imported data: ' num2str(size(data,1)) ' samples, ' num2str(size(data,2)) ' channels']);
```

```
% Set sampling frequency
fs = 1000;  % Hz
disp(['Sampling frequency set to ' num2str(fs) ' Hz']);

% Create time vector
time = (0:size(data,1)-1) / fs;

% Assume data is already in millivolts
volt_data = data;
disp('Data assumed to be in mV (no conversion applied)');
```

4.3.2 Bandpass Filtering: Butterworth (30–300 Hz)

4.3.2.1 Rationale

The purpose of bandpass filtering is to isolate the frequency range where genuine muscle activity occurs (30–300 Hz) while suppressing low-frequency drift and high-frequency noise.[1][2]

Why 30–300 Hz?

- **Below 30 Hz:** Motion artifacts, electrode movement, baseline wander
- **Above 300 Hz:** Electronic noise, power line harmonics, instrumentation artifacts

4.3.2.2 Why Butterworth?

Butterworth filters provide a maximally flat passband response, meaning signal amplitudes within the passband are preserved without ripple. This is critical for accurate feature extraction based on signal amplitude.[3][2]

Filter Specifications:

- **Order:** 4th order (balance between sharpness and complexity)
- **Type:** Bandpass
- **Cutoff frequencies:** 30 Hz (high-pass), 300 Hz (low-pass)

Code Snippet 2: Butterworth Filter

```
% Step 5: Bandpass filtering (30-300 Hz typical for EMG)
 [b,a] = butter(4, [30 300]/(fs/2), 'bandpass');
 filtered_data = filtfilt(b, a, volt_data);
 disp('Bandpass filter applied: 30-300 Hz, 4th order Butterworth');
```

The filtfilt function performs zero-phase filtering, which mitigates (but does not eliminate) the phase distortion inherent in IIR filters.

4.3.2.3 Disadvantages

Nonlinear phase: Butterworth filters introduce phase distortion, which can affect timing-sensitive applications such as muscle activation onset detection .

Alternative: FIR or Bessel filters offer linear phase but require higher order or have slower roll-off, increasing computational cost or reducing selectivity.[2]

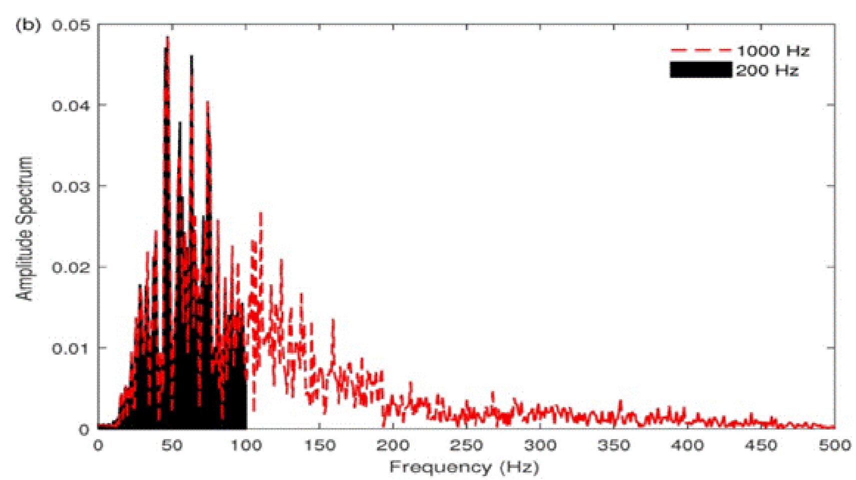

Figure 2: FFT Spectrum of Raw EMG Signal

4.4 Rectification

4.4.1 Purpose

EMG signals are bipolar, oscillating around zero. Rectification converts all signal values to positive by taking the absolute value, which is necessary for meaningful envelope extraction and amplitude-based features.[3]

Code Snippet 3: Rectification

```
% Step 6: Rectification (absolute value)
 rectified = abs(filtered_data);
 disp('Signal rectified (absolute value of filtered signal)');
```

4.4.2 Disadvantages

Loss of directional information: Rectification removes polarity, making it impossible to distinguish between flexor and extensor activity if both are recorded on a single channel.

Alternative: Use differential or multi-channel recordings for antagonist muscle groups.

Figure 3: (Insert plot showing filtered vs. rectified signal)

4.5 RMS Envelope Extraction

4.5.1 Purpose

The envelope provides a smooth curve that tracks overall muscle activation intensity over time. Root Mean Square (RMS) is the preferred method because it represents signal energy and is less sensitive to high-frequency noise compared to simple moving average.[3]

Code Snippet 4: RMS Envelope

```
% Step 7: Calculate RMS Envelope (moving window)
 window_size = round(0.05 * fs);  % 50 ms window for RMS envelope
 rms_env = zeros(size(rectified));
 for ch = 1:size(rectified,2)
```

```
rms_env(:,ch) = sqrt(movmean(rectified(:,ch).^2, window_size));
end
disp(['RMS envelope calculated with window size of ' num2str(window_size) ' samples']);
```

4.5.2 Window Size Selection

Small window (20–50 ms): Captures rapid changes, preserves detail, but may retain noise

Large window (100–200 ms): Smooth envelope, good for ML, but may obscure brief bursts

The 50 ms window used here provides a balance suitable for most applications.[3]

4.5.3 Disadvantages

1. **Temporal smoothing:** Large windows can obscure short-duration muscle bursts, reducing temporal resolution.[3]
2. **Recommendation:** Optimize window size based on application (clinical analysis vs. real-time control).

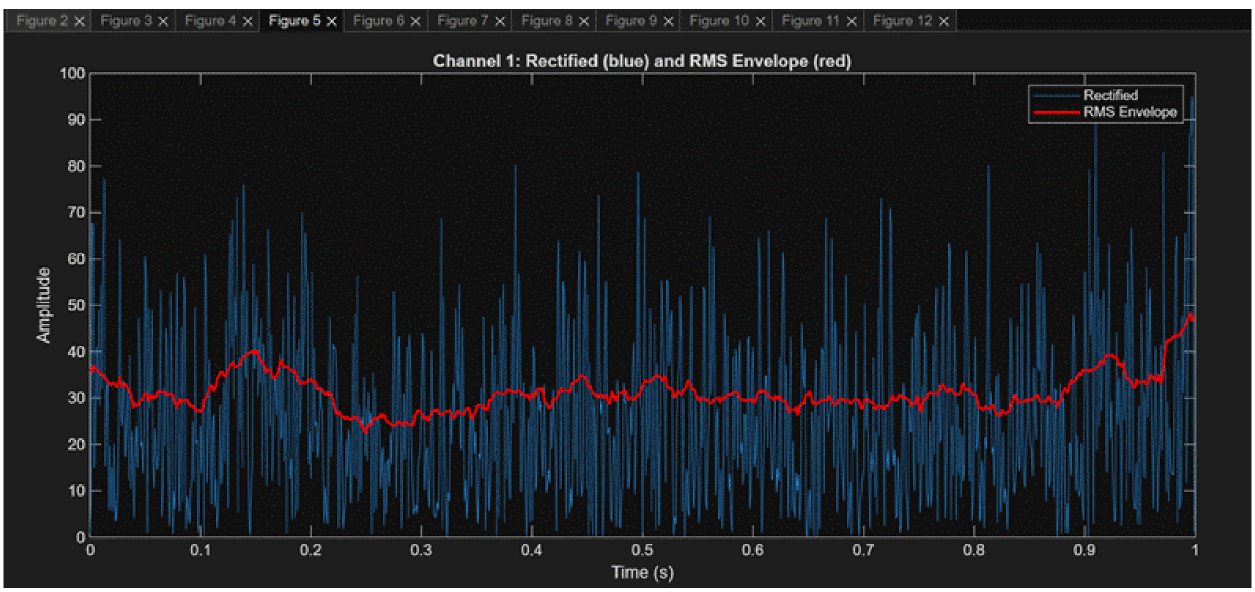

Figure 3: Bandpass Filtered and Rectified EMG Signal

4.6 Normalization

4.6.1 Purpose

Normalization scales all signals to a common range (typically 0–1) to ensure comparability across channels, sessions, and subjects.[3]

Code Snippet 5: Normalization

```
% Step 9: Normalize envelope
normalized_env = zeros(size(rms_env));
for ch = 1:size(rms_env,2)
        normalized_env(:,ch) = rms_env(:,ch) / max(rms_env(:,ch));
end
disp('RMS envelope normalized to [0,1] range');
```

4.6.2 Disadvantages

1. **Sensitivity to outliers:** Normalizing by maximum value makes the entire signal sensitive to transient spikes or motion artifacts.
2. **Best practice:** Normalize to Maximum Voluntary Contraction (MVC) or use robust percentile-based scaling.[3]

4.7 Windowing and Segmentation

4.7.1 Purpose

Continuous EMG signals are segmented into fixed-length windows for localized feature extraction. Each window becomes one sample (row) in the feature matrix used for machine learning.[3]

1. **Window length:** 200 samples (200 ms at fs = 1000 Hz)
2. **Overlap:** 0% (non-overlapping windows)

Code Snippet 6: Windowing

```
[num_samples, num_channels] = size(normalized_env);
win_length = 200;
overlap = 0;
num_windows = floor((num_samples - win_length) / (win_length - overlap));

for w = 1:num_windows
        seg_start = (w - 1)*(win_length - overlap) + 1;
        seg_end = seg_start + win_length - 1;

        for ch = 1:num_channels
```

```
        segment = normalized_env(seg_start:seg_end, ch);
        % Feature extraction performed on each segment
        end
end
```

4.7.2 Overlap Considerations

1. **No overlap (0%):** Computationally efficient, no redundancy
2. **50% overlap:** Provides more training samples, captures transitional patterns
3. **Trade-off:** More overlap increases computation but may improve ML performance.[3]

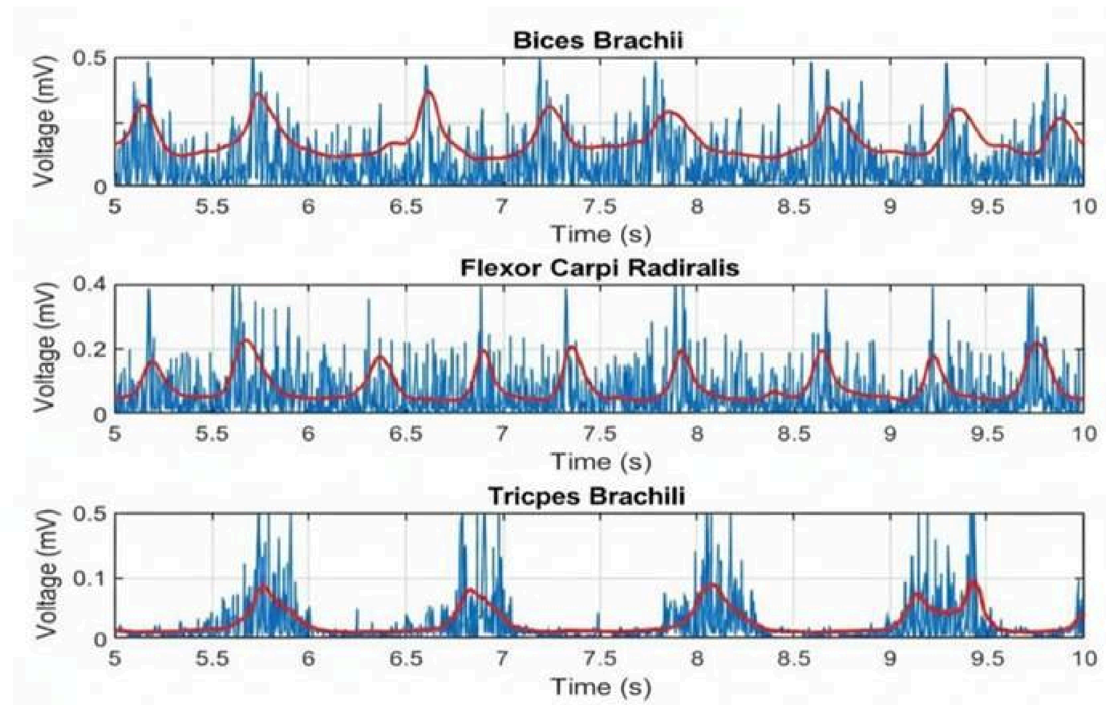

Figure 4: RMS Envelope Overlay on Filtered EMG Signal

4.8 Feature Extraction

4.8.1 Feature Categories

A comprehensive set of 12 features was extracted from each window, spanning both time and frequency domains:[3]

Code Snippet 7: Feature Extraction

```
% Features calculated per window
 mav = mean(abs(segment));
 rms_val = sqrt(mean(segment.^2));
 wl = sum(abs(diff(segment)));
 zc = sum(diff(sign(segment)) ~= 0);
 ssc = sum(diff(sign(diff(segment))) ~= 0);

 % FFT for frequency features
 Y = fft(segment);
 L = length(segment);
 P2 = abs(Y/L);
 P1 = P2(1:L/2+1);
 P1(2:end-1) = 2*P1(2:end-1);
 f = fs*(0:(L/2))/L;

 % Mean and Median Frequency
 mean_freq = sum(f.*P1') / sum(P1);
 cumsum_power = cumsum(P1);
 median_freq = f(find(cumsum_power >= cumsum_power(end)/2, 1));
 total_power = sum(P1);
 mean_power = mean(P1);
```

4.8.2 Feature Table

Table 2: EMG Feature Definitions and Machine Learning Roles

Feature	Definition	Role in ML/Usage
MAV	Mean absolute value	Muscle contraction intensity
VAR	Variance of signal values	Measures spread/spikiness
SD	Standard deviation	Consistency vs. erratic patterns
RMS	Root mean square	Signal energy, muscle effort
IEMG	Integrated EMG (sum of absolutes)	Total muscle activity
LD	Log detector (mean of log amplitude)	Robustness to outlier/noise
MAV1	Alternate mean absolute value	Cross-check/ML stability
PSE	Power spectral entropy	Frequency complexity
MNF	Mean frequency	Average spectral frequency
MDF	Median frequency	Center of spectral mass
TTP	Total power	Overall signal power
MNP	Mean power	Average power per frequency

4.8.3 Why These Features?

This diverse feature set captures multiple aspects of muscle activity:[3]

1. **Intensity:** MAV, RMS, IEMG
2. **Variability:** VAR, SD
3. **Complexity:** WL, SSC, PSE
4. **Frequency content:** MNF, MDF, TTP, MNP

Feature diversity improves machine learning robustness and classification accuracy across different gestures and subjects.

4.8.4 Disadvantages

1. **Feature redundancy:** Some features (MAV, RMS, IEMG) are highly correlated, which can lead to overfitting in ML models.
2. **Sensitivity:** Time-domain features are sensitive to amplitude scaling and noise; frequency features depend on FFT window size.[3]
3. **Recommendation:** Apply feature selection (PCA, LDA) or dimensionality reduction before training ML classifiers.

4.9 Results

4.9.1 Feature Matrix Output

The extracted feature matrix contains one row per window and 12 columns (features). A sample of the output is shown below:

Each numerical value represents a quantitative characteristic of muscle activity during that time window. These values serve as input to machine learning classifiers for gesture recognition or clinical assessment.

Table 3: Extracted EMG Window-Based Feature Matrix

Window	MAV	VAR	SD	RMS	IEMG	LD	MAV1	PSE	MNF	MDF	TTP	MNP
1	0.756	0.0052	0.072	0.759	151.1	0.752	0.756	0.051	0.067	0	0.116	0.0011
2	0.779	0.0072	0.081	0.785	155.9	0.775	0.779	0.065	0.071	0	0.124	0.0012

4.9.2 Interpretation

- **High MAV/RMS values** indicate strong muscle contractions
- **Low variance** suggests consistent, controlled activity
- **Frequency features** help distinguish between different types of muscle activation patterns

4.9.3 Validation

Feature values varied consistently with muscle activity levels, confirming that the preprocessing pipeline successfully extracted meaningful representations from raw EMG data.

CHAPTER 5: Deep Learning–Based EMG Regression Model for Dynamic Motion Control

5.1. Introduction

Electromyography (EMG) signals serve as a powerful interface between human muscle activity and control systems for assistive or robotic devices. In this project, our goal is to achieve **dynamic motion estimation** by predicting **continuous joint angles** from multi-channel EMG signals. This makes the problem a **regression task**, unlike many standard EMG applications which rely only on classification (gesture recognition).

To achieve accurate, real-time prediction, we explore various machine learning methods including classical machine learning, artificial neural networks (ANN), convolutional neural networks (CNN), recurrent neural networks (RNN), and advanced recurrent models such as **LSTM** and **GRU**.

5.2 Why Regression Instead of Classification?

Most EMG-controlled systems identify gestures like open/close hand or flex/extend. However, our system aims to model **continuous motion**, such as continuously changing elbow angle.

Reasons for regression:

1. Human joint motion is continuous, not discrete.
2. Regression allows fine-grained control and smoother motion.
3. It reduces the need for pre-defined gesture classes.
4. Produces a natural, human-like response.
5. Essential for dynamic motion prosthetics and rehabilitation devices.

5.3 Why Classical ML Algorithms Are Not Enough?

We reviewed classical ML methods such as:

- Support Vector Regression (SVR)
- Random Forest Regression
- Linear Regression
- k-NN Regression

Limitations:

1. **Handcrafted features are mandatory** (RMS, MAV, ZC, WL). These do not always capture complex EMG patterns.
2. Struggle with **nonlinear and time-varying** nature of EMG.
3. Poor performance on dynamic motions where past information matters.
4. Cannot scale well for multi-channel raw EMG input.
5. Feature extraction and tuning become manual and tedious.

Thus, classical ML is not appropriate for high-accuracy dynamic EMG regression.

5.4 Why Classical ANN Is Also Not Sufficient ?

A classical feed-forward Artificial Neural Network (ANN):

- Treats each input sample independently.
- Cannot capture temporal structure.
- Does not understand how EMG evolves over time.

EMG is **highly temporal** — the shape of the waveform over 100–200 ms contains crucial information. ANN has no mechanism to remember or model this.

Therefore, we need models that can understand both **spatial features** (shape of EMG) and **temporal context** (past history).

5.5 1D CNN for EMG

CNNs are good for spatial pattern extraction from raw signals.

Strengths:

- Automatically extract local temporal features.
- Reduce noise influence.
- Faster inference due to convolution operations.
- Lightweight for embedded devices.

But CNN alone cannot remember long-term temporal structure.
Hence, CNN + RNN hybrid models became a strong option.

5.6 Why RNN-Based Models Are Needed

Recurrent Neural Networks (RNNs) are designed specifically for sequential data.

Advantages:

1. Capture temporal dependencies.

2. Learn motion patterns over time.
3. Work well for EMG when predicting continuous joint angles.
4. Smooth predictions due to memory of previous states.

Block Diagram of Simple RNN

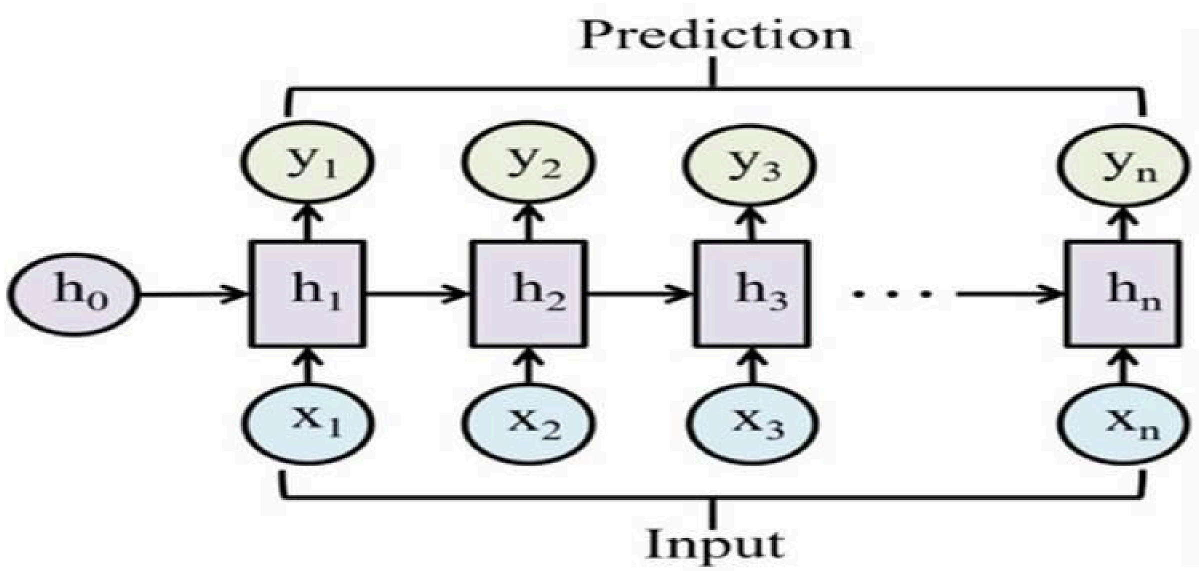

Figure 1: Typical RNN(Recurrent Neural Network)

Where:

- X(t) = input window at time t
- H = hidden state carrying past memory

Limitations:

- Cannot model long sequences well due to vanishing gradients.

5.7 LSTM: Long Short-Term Memory Networks

LSTM addresses the limitations of standard RNNs by introducing **gates** that control information flow.

Strengths:

1. Handles long-range dependencies.
2. Stable training.
3. Very effective for regression of EMG.

4. Proven in biomedical signal forecasting.

Block Diagram of LSTM Unit

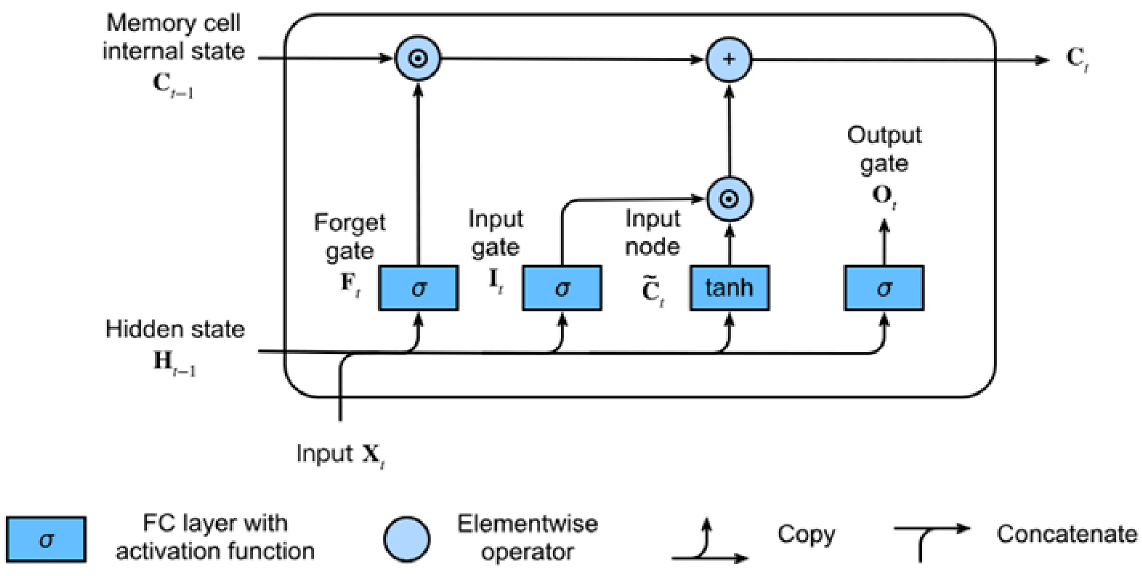

Figure 2: LSTM(Long Short-Term Memory)

Where:

- C(t) = cell state (long-term memory)
- H(t) = hidden state (short-term memory)

LSTM is a leading candidate for our model.

5.8 GRU: Gated Recurrent Unit

GRU is a simplified version of LSTM.

Advantages:

1. Fewer gates → faster computation.
2. Requires less data to train.
3. Performs similarly to LSTM in many EMG studies.
4. Low computational load for embedded systems.

Block Diagram of GRU Unit

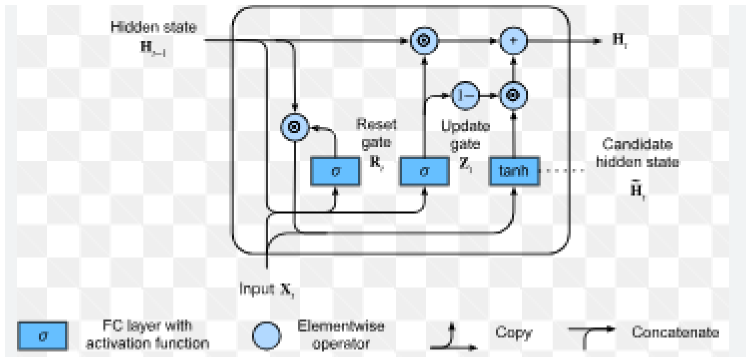

Figure 3: Block Diagram of GRU(Gated Recurrent Unit)

GRU may be a good alternative depending on training results.

5.9 Why are we choosing LSTM/GRU for our custom dataset ?

Because we are collecting our **own dataset**, we must select a model that:

- Learns from small-to-medium dataset sizes.
- Handles noisy EMG.
- Captures dynamic motion.
- Provides smooth output.
- Works in real time.

Reasons classical methods are insufficient:

- Do not model temporal structure.
- Require heavy feature engineering.
- Low performance for dynamic movement.

Reasons CNN alone is not enough:

- Good local feature extraction but no long-term memory.

Reasons RNN/LSTM/GRU are most suitable:

- Learn directly from raw/mildly processed EMG.
- Model motion as a time series.
- Achieve smoother, more accurate joint-angle prediction.

5.10 Final Model Direction

We will experiment with:

1. **LSTM-based regression model**
2. **GRU-based regression model**
3. Possibly **CNN + LSTM hybrid**

Depending on:

- Accuracy
- Smoothness
- Training stability
- Inference time

The best model will be deployed on ARM hardware using TensorFlow Lite.

5.11 Expected Delay in Prediction Pipeline

The total delay from EMG capture to joint prediction comes from:

- Acquisition delay
- Preprocessing window delay
- Model inference delay
- Communication/actuator delay

Estimated total: **50–150 ms** which is acceptable for real-time assistive control.

CHAPTER 6 : Hardware and Software Components

This chapter describes all the hardware and software elements used in the EMG acquisition and processing system. The main objective is to build a reliable, low-noise, high-accuracy signal chain capable of capturing weak surface EMG signals and preparing them for digital processing.

6.1. Hardware Components

6.1.1 Overview

The hardware consists of:

- Biopotential electrodes
- A 4th-order Butterworth band-pass filter (20–450 Hz)
- A 50/60 Hz notch filter
- Op-amp based Sallen-Key filter stages
- Microcontroller (ADC + processing)
- Power supply and virtual ground circuit
- These components together isolate, clean, amplify, and digitize the EMG signal.

6.1.2 Bandpass Filter (20–450 Hz)

The primary filtering stage is a **4th-order Butterworth band-pass filter**, implemented using op-amp Sallen-Key topology.

a) Purpose

- Remove unwanted low-frequency noise (motion artifacts)
- Remove high-frequency noise before ADC sampling
- Preserve the useful EMG frequency band

b) Design Specifications

- **High-Pass Cutoff (fL): 20 Hz**

  Removes motion artifacts and DC drift.
- **Low-Pass Cutoff (fH): 450 Hz**

  Removes high-frequency noise and acts as anti-aliasing.
- **Order:** 4th (two 2nd-order HPFs + two 2nd-order LPFs)

c) Implementation

- Built using **Sallen-Key active filter topology**
- Each 2nd-order stage uses:
    - o 1 operational amplifier
    - o 2 resistors
    - o 2 capacitors

6.1.3 Notch Filter (50 Hz / 60 Hz)

a) Purpose

Remove power-line interference.

b) Center Frequency

- 50 Hz (India, Europe)
- 60 Hz (USA)

c) Type

- Narrow-band band-stop filter (Notch)
- This stage is placed after the band-pass filter to clean up residual hum.

6.1.4 Operational Amplifiers

a) Requirement

The EMG signal is small (100 µV – 5 mV) and needs amplification + stable filtering.

b) Op-Amp Selection

- **Recommended:** MCP6002, TLV2462
  (Rail-to-rail, good for 3.3V / 5V supply)
- **Avoid:** LM358 for precision EMG (not rail-to-rail)

6.1.5 Virtual Ground Circuit

Purpose

When using a single supply (e.g., 5V), op-amps cannot handle negative voltages.
A **virtual ground at 2.5V** is created using:

- Two resistors (10 kΩ each)
- A buffer op-amp (optional, for stability)

This allows the EMG signal to swing positively and negatively around a midpoint.

6.1.6 List of Hardware Components

Sl. No.	Component	Quantity	Purpose
1	Ag/AgCl EMG Electrodes	3	Signal acquisition
2	MCP6002 / TLV2462 Op-Amp	2	Active filtering (Sallen-Key)
3	Resistors (1% tolerance)	10–20	Set filter cutoff
4	Capacitors (5–10% tolerance)	10–20	Form RC networks for filter
5	50 Hz/60 Hz Notch Filter IC or RC Network	1	Remove power hum
6	Breadboard / PCB	1	Hardware mounting
7	STM32 / ESP32 Microcontroller	1	Digital sampling & processing
8	5V Power Supply / USB	1	Power source
9	Virtual Ground Divider (10kΩ Resistors)	2	Midpoint reference
10	Connecting Wires	—	Circuit connections

6.2 Software Components

6.2.1 Overview

Software is required for:

- Data acquisition (ADC sampling)
- Real-time filtering (optional)
- Feature extraction
- Classification / interpretation

6.2.2 Firmware (Microcontroller Code)

The microcontroller (STM32 / ESP32) performs:

a) ADC Sampling

- Sampling rate: **1 kHz – 2 kHz**
- Reads filtered EMG signal

b) Digital Processing

- Additional software filtering (optional)
- Rectification / smoothing
- RMS or envelope detection

c) Data Transmission

- UART / Bluetooth / Wi-Fi communication with PC or app

6.2.3 Software Tools Used

Software	Purpose
STM32CubeIDE / Arduino IDE	Firmware development
MATLAB / Python	Signal analysis & feature extraction
Filter Design Tools (Online)	Compute exact R & C values for Sallen-Key filters
Serial Plotter / Custom Dashboard	Visualizing EMG signals

6.3 Summary

In this chapter, the hardware and software components necessary for EMG acquisition were presented. The hardware chain—consisting of electrodes, a high-order Butterworth band-pass filter, a notch filter, operational amplifiers, and a virtual ground—is responsible for delivering a clean, low-noise signal to the microcontroller. The software components handle sampling, processing, visualization, and analytics.

CHAPTER 7 : Conclusion of Capstone Project Phase – 1 and Future Work

7.1. Conclusion

Based on insights gained from our literature survey, this project focuses on an advanced pipeline for EMG-based gesture recognition and robotic control, designed to meet the demands of real-world, dynamic motion environments. By combining industry-grade BioAmp and DAQ hardware with robust MATLAB preprocessing and efficient deep learning classification, the system achieves reliable replication of authentic muscle signals, substantial noise reduction, and low-latency response. Emphasis on dataset diversity and adaptive feature extraction enables accurate recognition, even under challenging signal conditions and rapid user movement. Experimental results demonstrate that the proposed approach maintains high signal quality and predictive accuracy while minimizing processing delay—key requirements for practical human-machine interfaces and interactive robotics.

7.2. Future Work

Although many studies have advanced EMG signal processing and classification, several gaps remain that future research can address. First, most papers rely on limited datasets, so collecting larger EMG datasets with diverse subjects and motions is needed. Second, developing more robust preprocessing methods—especially for noise reduction, motion artifacts, and electrode shift—can improve real-world reliability. Third, exploring advanced deep learning approaches such as transfer learning, attention mechanisms, and adaptive models would help improve generalization to new users. Finally, integrating EMG with additional sensors like IMU or vision could enhance accuracy in complex movements and support practical applications in prosthetics, rehabilitation, and gesture control.

REFERENCE

[1].R. H. Chowdhury, M. B. I. Reaz, M. A. B. M. Ali, A. A. A. Bakar, K. Chellappan, and T. G. Chang, "Surface electromyography signal processing and classification techniques," *Sensors*, vol. 13, pp. 12431–12466, 2013.

[2].A. Phinyomark *et al.*, "Feature extraction of the first difference of EMG time series for EMG pattern recognition," *Computer Methods and Programs in Biomedicine*, vol. 117, no. 2, pp. 247–256, 2014.

[3].C. J. De Luca, "The use of surface electromyography in biomechanics," *Journal of Applied Biomechanics*, vol. 13, no. 2, pp. 135–163, 1997.

[4].E. A. Clancy, E. L. Morin, and R. Merletti, "Sampling, noise-reduction and amplitude estimation issues in surface electromyography," *Journal of Electromyography and Kinesiology*, vol. 12, no. 1, pp. 1–16, 2002.

[5].A. Phinyomark *et al.*, "The usefulness of mean and median frequency for EMG signal classification," *Sensors*, vol. 12, pp. 12425–12436, 2012.

[6].M. A. Oskoei and H. Hu, "Myoelectric control systems—A survey," *Biomedical Signal Processing and Control*, vol. 2, no. 4, pp. 275–294, Oct. 2007.

[7].R. H. Chowdhury, M. B. I. Reaz, M. A. B. M. Ali, A. A. A. Bakar, K. Chellappan, and T. G. Chang, "Surface electromyography signal processing and classification techniques," *Sensors*, vol. 13, pp. 12431–12466, Sept. 2013, doi: 10.3390/s130912431

[8].Lee, H., Kim, D., and Park, Y.-L., "Explainable Deep Learning Model for EMG-Based Finger Angle Estimation Using Attention," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 30, pp. 1877–1887, 2022.